Homework 1

2.1.1.a Let μ be a positive inner regular locally finite measure. Suppose there exists a constant D such that for all $x \in \mathbb{R}^n$ and $r > 0$ we have

$$\mu(3B(x,r)) \leq D\mu(B(x,r)).$$

Define the weak $L^{1,\infty}$ norm $\|\cdot\|_{L^{1,\infty}}$ by

$$\|f\|_{L^{1,\infty}} = \sup_{\lambda>0} \lambda\mu(|f| > \lambda).$$

We will show the uncentered maximal function M is continuous $L^1(\mathbb{R}^n, \mu) \to L^{1,\infty}(\mathbb{R}^n, \mu)$. Define $E_\alpha = \{x \in \mathbb{R}^n : Mf(x) > \alpha\}$. We need to show

$$\|Mf\|_{L^{1,\infty}} \leq C\|f\|_{L^1},$$

and it will suffice to show for some $\alpha > 0$ that

$$\alpha\mu(E_\alpha) \leq C\|f\|_{L^1}.$$

First, I claim E_α is open. For $x \in E_\alpha$, there must be a ball B_x containing x such that the average value of f on B_x is greater than α. Then this ball is in fact contained in E_α, since for any point $y \in B_x$ we can choose a ball $B_y \subset B_x$ containing y, and this proves $Mf(y) > \alpha$.

Now use inner regularity. Choose any compact $K \subset E_\alpha$ and cover K in balls $\{B_x\}_{x\in K}$ such the average value of f on every B_x is greater than α. Choose a finite subcover $\{B_i\}_1^N$ and choose a further subcover $\{B_{i_k}\}_1^M$ such that $\cup B_i \subset \cup 3B_{i_k}$. Then

$$\mu(K) \leq \mu\left(\bigcup_{k=1}^M 3B_{i_k}\right) \leq D\sum \mu(B_{i_k}) \leq \frac{D}{\alpha}\sum \int_{B_{i_k}} |f(y)|dy \leq \frac{D}{\alpha}\|f\|_{L^1}.$$

2.1.2.a Let $\mathscr{F}$ be a finite family of open intervals in $\mathbb{R}$. We prove there exist two subfamilies $\mathscr{F}_1$ and $\mathscr{F}_2$ of $\mathscr{F}$, each family pairwise disjoint, such that the union of the sets in both families is equal to the union of $\mathscr{F}$.

Without loss of generality, we can reduce to the case when $\cup\mathscr{F}$ is connected, and we can remove any intervals strictly contained in larger ones. If the union is not disjoint, we can just solve the problem on disjoint pieces and combine the individual families in any way.

We proceed by induction. The case of $n = 1$ is trivial. Now assume we have proven that for any family of $k - 1$ elements, we can find two disjoint subfamilies as above.

Let $\mathscr{F}$ be a connected family of k intervals. Remove the leftmost set U_0, which must exist since $\mathscr{F}$ is finite, and we have removed any sets contained in others. Then from $\mathscr{F} \setminus \{U_0\}$, choose subfamilies $\mathscr{G}_i$, $i = 1, 2$ which satisfy the desired property.